

Numerical modeling of coupled phenomena in a mechanically stirred molten-glass bath heated by induction

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Abstract

A mechanically stirred molten-glass bath heated by direct induction in a cold crucible was numerically modeled. The aim of the study was to develop numerical tools to understand thermal, hydrodynamic and electromagnetic phenomena that occur in the bath. Models and coupling between these phenomena are described. This coupling and the high content of elements in the 3D mesh result in a long calculation time. The study demonstrates how to couple programs to yield the highest degree of accuracy in the shortest calculation time possible. Numerical studies are also used to characterize the fluid dynamic behavior and heat transfer in an industrial-size tank. Classical correlations commonly used to characterize stirrer efficiency and heat transfer for fluids with constant physical properties were adapted for molten glass. The power number N_p and the Nusselt number Nu are used as macroscopic indicators. The results of these global studies will be useful for the operation and optimization of the vitrification facilities.

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1. Introduction

Since the late 1950s, many research and development programs have focused on the problem of nuclear waste reprocessing and containment. Vitrification is suitable for safe long-term conditioning of high-level radioactive waste. The process involves incorporation of the ultimate waste in molten glass. Waste results from spent-fuel reprocessing and is composed of minor actinides and fission products. The amorphous structure of glass makes it possible to contain such waste by integration of all the elements of fission-product solutions within the matrix at the atomic scale, resulting in true containment rather than encapsulation. To ensure the long-term stability of the waste package, the chemical, mechanical and thermal properties of the glass must not change with time. A continuous two-step vitrification process was set up by AREVA NC on six vitrification lines at La Hague (France) in 1989. This process was developed at the Marcoule Vitrification Unit starting in 1978. Over the

past 20 years, the French Atomic Energy Commission (CEA, France) has been developing a new vitrification process to meet requirements including the vitrification of other types of waste and a reduction in waste volumes. The technology is under investigation at Marcoule and is intended to replace metal pots on one of the six vitrification lines of the reprocessing plant at La Hague. Features of the process include an inductor supplied by a sinusoidal current and cooling of all the walls (Fig. 1). For a cold crucible with a diameter of 650 mm, the inductor surrounding the crucible is connected to a high-frequency current generator with output of 400 kW at a frequency of 300 kHz. The process is based on direct induction of electric currents in the glass. Energy dissipated by the induced currents due to the Joule effect keeps the glass molten. To allow the induction of electric currents, the crucible is divided into segments. The advantage of cooling the walls is mainly related to the formation of a thin layer of solidified glass, which forms a skull melter that insulates the walls from the molten glass. Thus, the walls are not corroded by the molten glass and in turn the molten glass is not contaminated. A cooled mechanical stirrer is added to thermally and chemically homogenize the molten glass. Large

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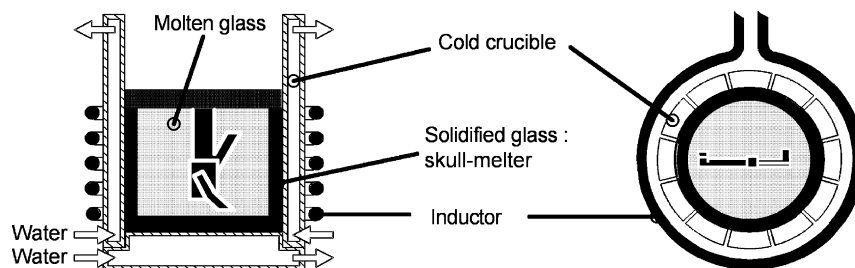


Fig. 1. Cold crucible vitrification process with the stirrer studied.

efforts are being made to identify the optimal process settings, which involves many fields, including electrical engineering, electromagnetism, hydrodynamics, heat transfer and chemical science. Hence, modeling is very useful in understanding and predicting phenomena that occur within the glass bath. In the present study, we consider electromagnetic, thermal and hydrodynamic phenomena within the molten-glass bath. The main difficulties encountered are related to two complications: (1) large thermal variations in the physical properties of the glass, which requires coupling of the electromagnetic, hydrodynamic and thermal phenomena; and (2) asymmetry created by the stirring systems, which requires 3D calculations. To date, only a few studies have been devoted to identifying numerical solutions for this process. To the best of our knowledge, there are no reports of 3D calculations to study mechanical stirring within a molten-glass bath heated by induction.

Some 2D studies have concentrated on the modeling of natural convection in an unstirred molten-glass bath heated by induction. In their studies, [Servant et al. \(1992\)](#) and [Saumabere \(1994\)](#) did not take fluid flow into account, and only thermal and electromagnetic phenomena were coupled. [Schiff et al. \(2000\)](#) studied all three aspects, but their electromagnetic model neglected border effects. [Lopukh et al. \(2004\)](#) did not consider coupling between the different phenomena: the maximum Joule power density was injected close to the crucible walls. Similar results will be obtained if the electrical conductivity is assumed to be constant.

A considerable number of studies have considered mechanical stirring. In fact, stirred tanks are widely used to carry out a variety of operations in chemical engineering. Background information and references are provided by [Brucato et al. \(1998\)](#). Many experimental investigations of hydrodynamics in vessels of different geometrical configurations agitated by various types of impeller have been carried out. These studies have been complemented by the application of advanced numerical modeling techniques to provide more detailed insight into such processes in a cost-effective way. Investigations to quantify stirred tank performance and heat transfer on the walls have also been carried out. The power number N_p and the Nusselt number Nu are commonly used as macroscopic indicators to characterize stirred systems. To date, most reports have regarded fluids with constant physical properties or non-Newtonian fluids (see, for instance, reviews by [Nagata, 1975](#); [Mohan et al., 1992](#)). No studies on macroscopic indicators within a glass bath or fluids

with large thermal variations in viscosity seem to have been performed. The difficulty is in choosing a viscosity value to compute the dimensionless numbers. In addition, few studies have been devoted to simulation of mechanical stirring within a molten-glass bath. The TNO Glass Group ([Op den Camp and Aume, 1998](#); [Op den Camp et al., 2002](#); [Hegen et al., 2002](#); [Novak and Kasa, 2005](#)) has developed software for 3D simulation of a TV panel production line. In these studies, the refining zone and spout are equipped with non-cooled stirrers. However, the glass bath is not heated by direct induction. Experimental temperature distributions or profiles calculated by an external program are used as thermal boundary condition. Thus, coupling was not developed.

The aim of the present study was to simulate electromagnetic, hydrodynamic and thermal phenomena during cold crucible vitrification. The goal is to take into account the mechanical stirrer and to develop coupling between these aspects. The coupling and the high amount of elements in the 3D mesh result in high calculation time. Thus, this present study shows how to couple the programs with highest degree of accuracy and with smallest calculation time possible. Secondly, classical correlations commonly used for fluids with constant physical properties to characterize stirrer efficiency and heat transfer have been adapted to the glass.

2. Mathematical models

2.1. Thermohydrodynamic model

Thermohydrodynamic calculations are based on the following assumptions:

- Only the glass bath is modeled. The flow is assumed to be laminar, 3D and unsteady.
- The fluid is assumed to be Newtonian.
- The Boussinesq assumption is used to model buoyancy. This model treats density as a constant value in all solved equations, except for the buoyancy term in the momentum equation, where ρ is defined as $\rho = \rho_0(1 - \beta(T - T_0))$, ρ_0 is the constant density of the flow ($\rho = 2800 \text{ kg m}^{-3}$), T_0 is the reference temperature ($T_0 = 293 \text{ K}$) and β is the thermal expansion coefficient ($\beta = 3 \times 10^{-5} \text{ K}^{-1}$). This approximation is accurate because changes in actual density are small; specifically, the Boussinesq approximation is valid because $\beta(T - T_0) \ll 1$.

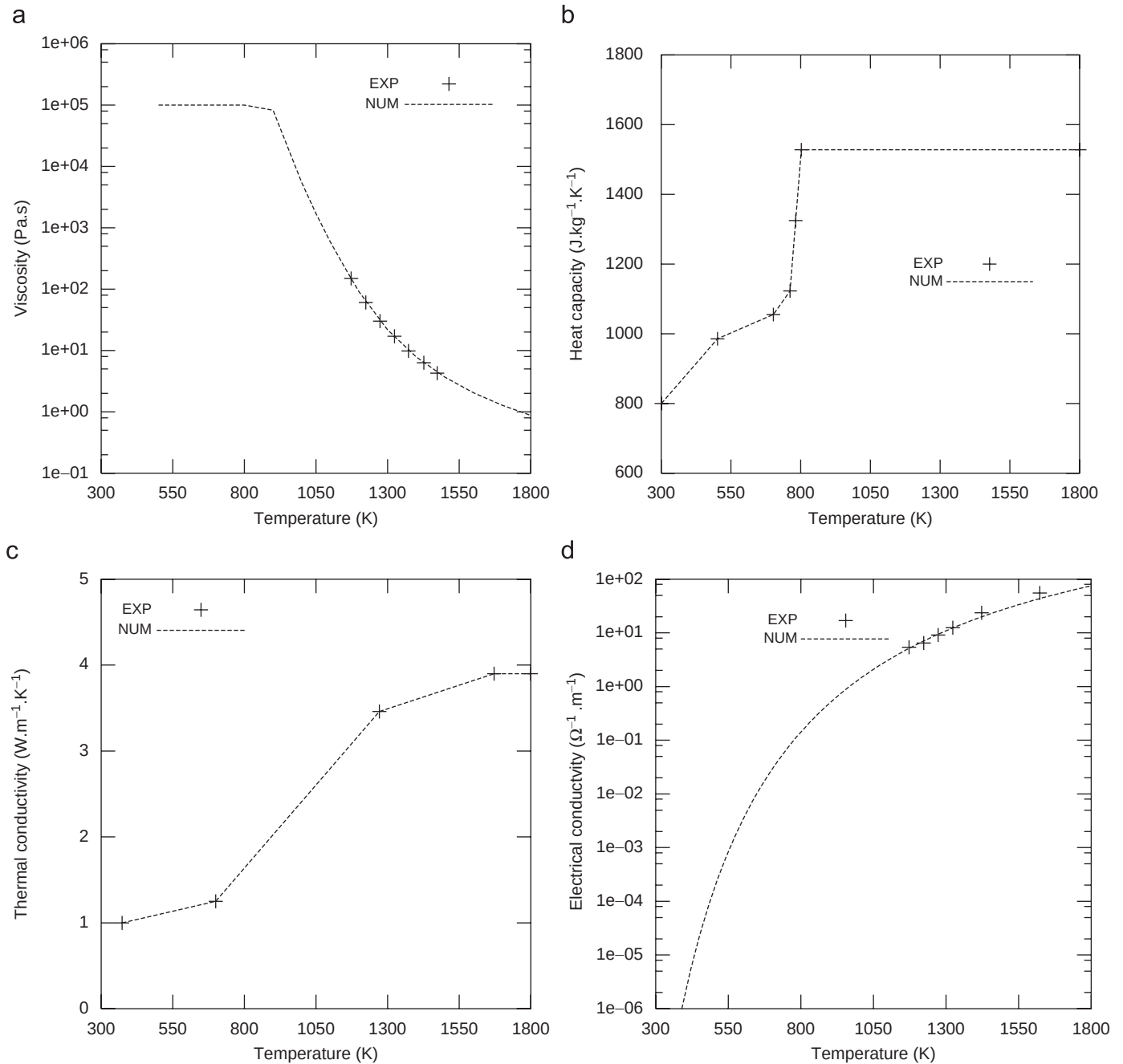


Fig. 2. Temperature dependence of: (a) viscosity; (b) heat capacity; (c) thermal conductivity; and (d) electrical conductivity. NUM, numerical laws; EXP, measurements carried out at Marcoule CEA (France).

- The physical properties are defined by temperature-fitted dependence functions. These laws are shown in Fig. 2.
- Lorentz forces due to induced currents and the electromagnetic field in the glass are neglected. The magnitudes of these forces are low compared to the forces generated by mechanical stirring (Roscini, 1990).
- Internal radiation in the glass bath is neglected; the modeled glass is almost completely non-transparent, even for thin layers.

The impeller blades sweep the domain periodically, and there are no stators or baffles. Consequently, it is possible to perform

calculations using a coordinate system that moves with the rotating equipment. The equations of motion are solved in a non-inertial frame, and the acceleration of the fluid is therefore augmented by additional terms that appear in the momentum equation. With the above assumptions, continuity (1), Navier Stokes (2), and thermal (3) equations may be formulated in the rotating frame as

$$\vec{\nabla} \cdot \vec{v}_r = 0, \quad (1)$$

$$\begin{aligned} \rho_0 \frac{\partial \vec{v}_r}{\partial t} + \rho_0 \vec{\nabla} \cdot (\vec{v}_r \vec{v}_r) + \rho_0 (2\vec{\Omega} \wedge \vec{v}_r + \vec{\Omega} \wedge \vec{\Omega} \wedge \vec{r}) \\ = -\vec{\nabla} p + \vec{\nabla} (\mu \vec{\nabla} \vec{v}_r) - \rho_0 \beta (T - T_0) \vec{g}, \end{aligned} \quad (2)$$

Table 1
Variables used to describe the boundary conditions

h_s ($\text{W m}^{-2} \text{K}^{-1}$)	T_c (K)	T_a (K)	T_{ar} (K)	ε (–)
20	333	303	423	0.9

$$\rho_0 \frac{\partial}{\partial t} (c_p T) + \rho_0 \vec{\nabla} \cdot (c_p T \vec{v}_r) = \vec{\nabla} \cdot (\lambda \vec{\nabla} T) + Q_{th}, \quad (3)$$

where $\vec{v}_r$ is the relative velocity in the rotating frame, $\vec{\Omega}$ is the angular velocity vector (rad s^{-1}), p is the pressure (Pa), μ is the dynamic viscosity (Pa s), $\vec{g}$ is the gravity vector (m s^{-2}), c_p is the heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$), and λ is the thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$). The source term Q_{th} in Eq. (3) represents the density of the Joule power (W m^{-3}), which is calculated by an electromagnetic program according to the relation:

$$Q_{th} = \frac{|j|^2}{2\sigma}, \quad (4)$$

where $|j|$ denotes the modulus of the current density (A m^{-2}) and σ is the electrical conductivity of the glass studied ($\Omega^{-1} \text{m}^{-1}$). The velocity boundary conditions are chosen to correspond to the physical model, i.e. no-slip boundary conditions along solid boundaries and a slip boundary condition on top of the glass bath. For the thermal boundary conditions, radiative and convective heat transfer is taken into account at the free surface. The heat flux through the free surface is calculated as

$$\varphi = \varepsilon \sigma_{sb} (T_s^4 - T_{ar}^4) + h_s (T_s - T_a), \quad (5)$$

where ε denotes the emissivity of the free surface glass, σ_{sb} is the Stefan-Boltzmann constant ($\sigma_{sb} = 5.67 \times 10^{-8} \text{W m}^{-2} \text{K}^{-4}$), T_{ar} is the reference temperature for radiative transfer, T_s is the free surface temperature, h_s is the external heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$), and T_a is the reference temperature for convective transfer. Heat losses through the walls are expressed according to

$$\varphi = h_w (T_w - T_c), \quad (6)$$

where h_w denotes the heat transfer coefficient for the wall, T_w is the glass temperature at the boundary, and T_c is the cooling water temperature. Only the glass bath is modeled, and thus h_w includes contact resistance between the glass and the cooled walls, conduction resistance inside the metal walls and, convection resistance between the walls and the cooling water.

The values of the variables used to define the thermal boundary conditions are summarized in Table 1. The method used to determine the values of h_w is presented in Section 4.1.

The thermohydrodynamic aspects were modeled using FLUENT CFD software and its accompanying mesh generation software, Gambit. The mesh is composed of 10^6 elements. It mainly includes tetrahedral and hexahedral elements. Local refinement of the mesh is carried out close to the cooled walls (i.e. where thermal gradients are high). The governing equations are solved sequentially using a finite-volume discretization process. In this segregated method, each discrete

equation is linearized implicitly with respect to the corresponding dependent variable. For the convection terms of each equation (momentum and energy) the third-order monotone upstream-centered schemes for conservation laws (MUSCL) scheme is used. The diffusion terms are central-differenced and second-order accurate. The pressure terms was interpolated using the pressure staggering option (PRESTO) scheme. The semi-implicit method for pressure-linked equations consistent (SIMPLEC) algorithm was used to introduce pressure into the continuity equation.

2.2. Electromagnetic model

The present impeller leads to a non-axisymmetric distribution of induced currents and temperature inside the glass. However, the fields are distorted only near the stirrer. Perturbation of the stirrer is limited in space. Thus, electromagnetic calculations are achieved using 2D-axisymmetric models within the glass bath.

A relevant characteristic scale that follows from sinusoidal variations of the electromagnetic field is the electromagnetic skin depth δ (m). This length, defined by Eq. (7), represents the diffusion depth of the electromagnetic field in the material at pulsation ω (s^{-1})

$$\delta = \sqrt{\frac{2}{\mu_0 \sigma \omega}}, \quad (7)$$

where μ_0 denotes the magnetic permeability ($\mu_0 = 4\pi \times 10^{-7} \text{H m}^{-1}$). For the conductor elements (crucible, coil, and base) the skin depth is very small, and thus the mesh is limited to the surface and a classical decreasing exponential law is used to calculate the electromagnetic variables within the skin depth (8)

$$\hat{f} = \hat{f}_s e^{-(1+i)n/\delta}, \quad (8)$$

where $\hat{f}$ denotes the complex magnitude of either the electric current density j , or the magnetic vector potential A (Wb m^{-1}), $\hat{f}_s$ is the value of $\hat{f}$ at the surface, and n is the wall-normal distance. Within the molten glass, the skin depth is much thicker and a volume mesh is therefore created.

In this calculation, the coil and base are assumed to be axisymmetric. The modeled cold crucible is 3D and periodic in the azimuthal direction. Thus, only one water-cooled segment of the cold crucible is meshed. The electromagnetic mesh is shown in Fig. 3a. The direction of the electric currents at the surface of the cold crucible is indicated in Fig. 3b.

The problem is defined by Eqs. (9), (10), and (11), where the scalar potential V (V) and the electric current density $\vec{j}$ are the unknowns. The vector potential $\vec{A}$ is given by the Biot–Savart law (12). The temperature dependence of electrical conductivity is shown in Fig. 2.

$$\vec{\nabla} \cdot \vec{j} = 0, \quad (9)$$

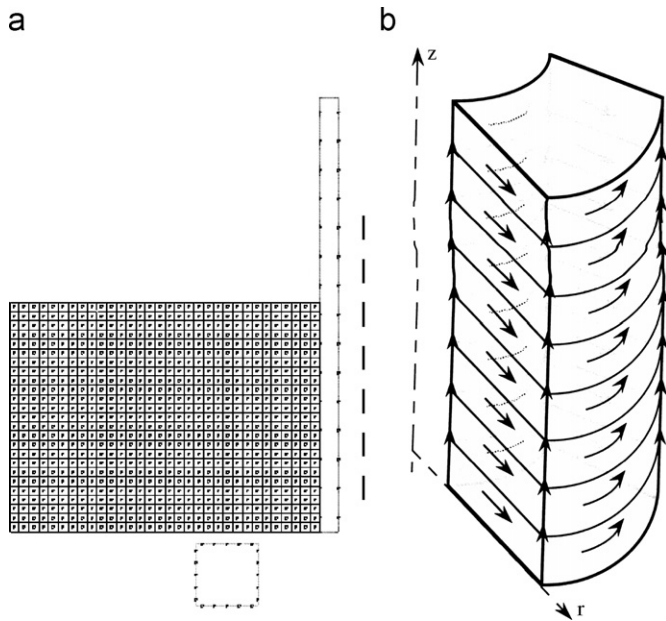


Fig. 3. Electromagnetic calculation. (a) Mesh of the elements; and (b) direction of the electric currents at the surface of the cold crucible.

$$\oint \vec{\nabla} V \cdot d\vec{l} = 0, \quad (10)$$

$$\vec{j} = -\sigma \left(\vec{\nabla} V + \frac{\partial \vec{A}}{\partial t} \right), \quad (11)$$

$$\vec{A}(M) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}}{r} dv. \quad (12)$$

The electromagnetic model is based on the OPHELIE program developed by the EPM-Madylam laboratory (Gagnoud and Leclercq, 1988). This program uses integral methods to solve the problem. The main advantages of the method are the possibility of using a surface mesh and of solving equations with only two unknowns: one component of the current density vector $\vec{j}$ and the scalar potential V .

3. Coupling

Analysis shows that the phenomena are highly coupled because the physical properties of the glass vary greatly with temperature (Fig. 2). Coupled modeling is therefore necessary. Coupling is only applied to calculations for the glass bath. The general coupling algorithm is detailed in Fig. 4.

An in-house supervisor program was developed to achieve the coupling. It successively and automatically runs the FLUENT and OPHELIE programs. The convergence criterion of each program is chosen by the user. The coupling iteration is controlled by a convergence criterion based on electrical conductivity changes between the last two iterates of the solution. The electrical conductivity and Joule power density are used to couple electromagnetism and thermohydrodynamics. As the electrical conductivity varies greatly with temperature, this variable is under-relaxed to avoid divergence of the solution.

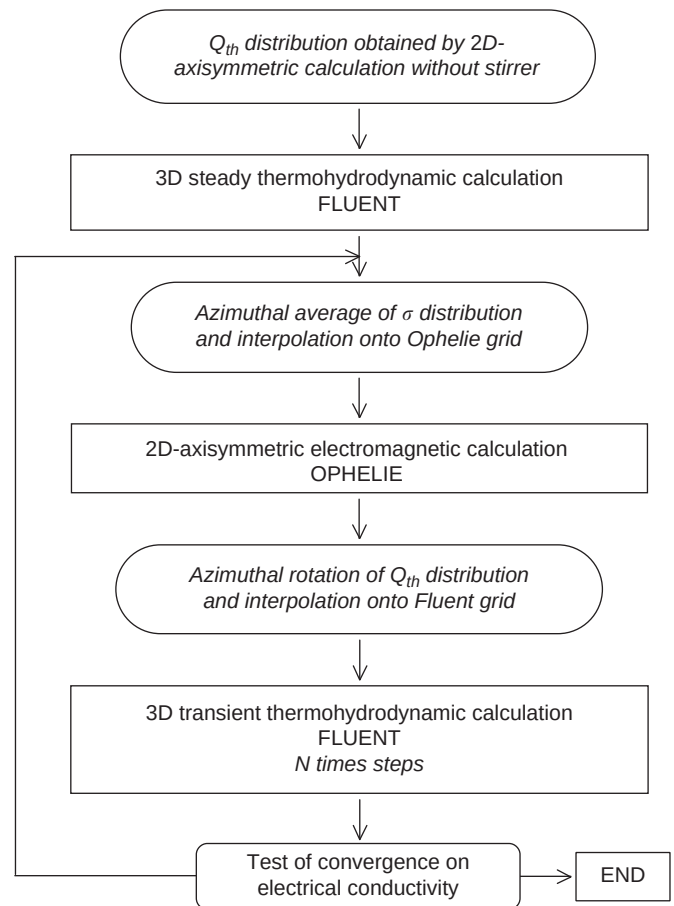


Fig. 4. Algorithm for coupling.

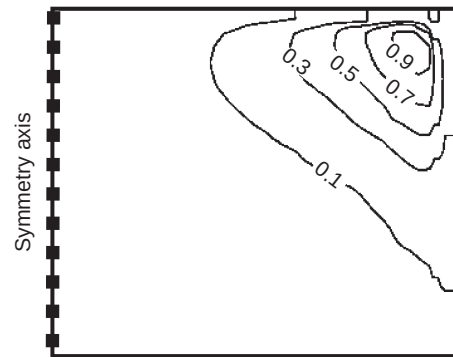


Fig. 5. Dimensionless axisymmetric power source distribution obtained by 2D-axisymmetric steady coupled calculation before coupling (i.e. without stirrer). $Q_{th \max} = 1.28 \times 10^7 \text{ W m}^{-3}$. $N = 10 \text{ rev min}^{-1}$. $P_{\text{tot}} = 60 \text{ kW}$.

Table 2

Calculation times for the coupling steps

3D thermohydrodynamic calculation 1 time step	10 min
2D-axisymmetric electromagnetic calculation	10 min
Interpolation steps	15 min
1 cycle of coupling	35 min

Coupling is performed within the loop between 3D thermohydrodynamic results and 2D electromagnetic results. Thus, additional steps have to be carried out during the coupling: the

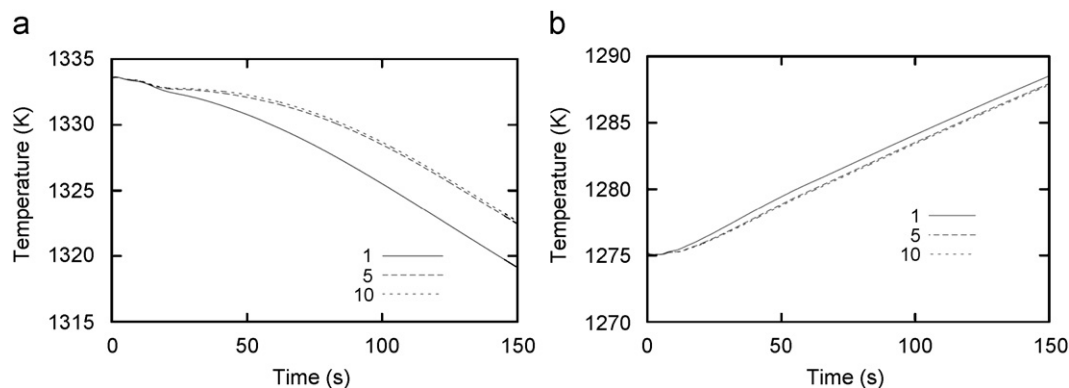


Fig. 6. Comparison of temperature obtained with 1, 5, and 10 thermohydrodynamic time steps at monitoring points A and B (positions specified in Fig. 8).

axisymmetric Joule power distribution is 3D-transformed by rotating about the vertical axis, and the 3D distribution of electrical conductivity is averaged to obtain an axisymmetric distribution. This study is specific because quasi-steady and transient calculations are coupled. The choice of transient thermohydrodynamic calculation is linked to the difficult convergence of a steady calculation. In fact, it is difficult to provide an initial solution that will allow the desired final solution to be obtained. Moreover, a steady calculation for which the sum of the heat fluxes on the boundaries is a negligible imbalance is used as an initial condition for the transient calculation to reduce calculation time. The steady calculation is performed using the axisymmetric Joule power distribution obtained from the 2D axisymmetric coupled calculation without considering the stirrer (Fig. 5).

Table 2 summarizes the calculation time for the coupling steps. One coupling cycle requires 35 min. Consequently, 440 days of computation would be necessary to run calculations for 1 h of experimental data with a thermohydrodynamic time step of 0.2 s. Thus, the goal is to optimize the number of thermohydrodynamic time steps to reduce the calculation time. Three cases have been tested: coupling was performed for 1, 5, or 10 thermohydrodynamic time steps between each coupling iteration. Comparison of the calculations obtained for these three cases shows very close results (Fig. 6). This can be explained by the large thermal diffusion time within the glass bath. Consequently, it is possible to use 10 time steps while keeping the accuracy high and reducing the calculation time by two-thirds.

4. Results

In this section, we present local temperature and velocity distributions and examine the macroscopic indicators of this stirred system.

4.1. Local study

Numerical results presented in this section were normalized by the maximum numerical value of the variable.

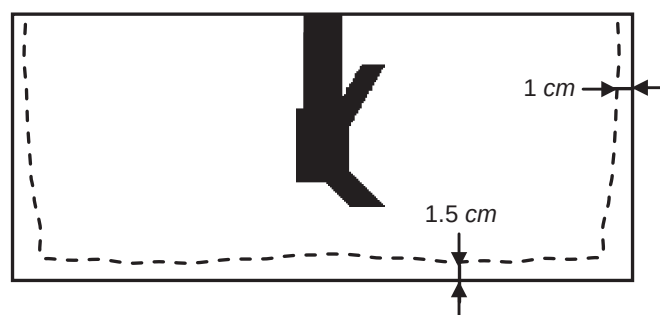


Fig. 7. Shape of the skull layer obtained after coupling convergence, $t=1000$ s.

The total Joule power injected within the glass bath is 60 kW and the stirrer is rotating at an angular velocity of 10 rev min^{-1} .

The first step involves choosing h_w defined in Eq. (6). This coefficient depends on the characteristics of the cooled walls: wall thickness, velocity, and physical properties of the cooling water and of the cooled walls. It is difficult to compute h_w using an analytical method because of the complex geometry of the cooled walls. Thus, we tested the sensitivity of the results to the heat transfer coefficient. The coefficients tested vary from 100 to $3000 \text{ W m}^{-2} \text{ K}^{-1}$. The highest value corresponds to the possible maximal value by considering only resistance due to the cooled walls: $h_{w \text{ max}} = \lambda_w / e$, where e denotes the thickness of the wall and λ_w is the thermal conductivity of the wall. The temperature and velocity distributions are similar within the glass bath when the heat transfer coefficient of the non-moving walls is greater than $300 \text{ W m}^{-2} \text{ K}^{-1}$ ($2500 \text{ W m}^{-2} \text{ K}^{-1}$ for the stirrer). This can be explained by the formation of a thin layer of solidified glass when the heat transfer coefficients of the non-moving and rotating walls are greater than 300 and $2500 \text{ W m}^{-2} \text{ K}^{-1}$, respectively. In fact, this solid zone between the molten glass and the cooled walls creates a resistance greater than that imposed by the heat transfer coefficient, and thus the influence of the heat transfer coefficients can be neglected. Consequently, we chose a value of $300 \text{ W m}^{-2} \text{ K}^{-1}$ ($2500 \text{ W m}^{-2} \text{ K}^{-1}$) for the non-moving walls (stirrer) in the present calculation. The solidified layer observed

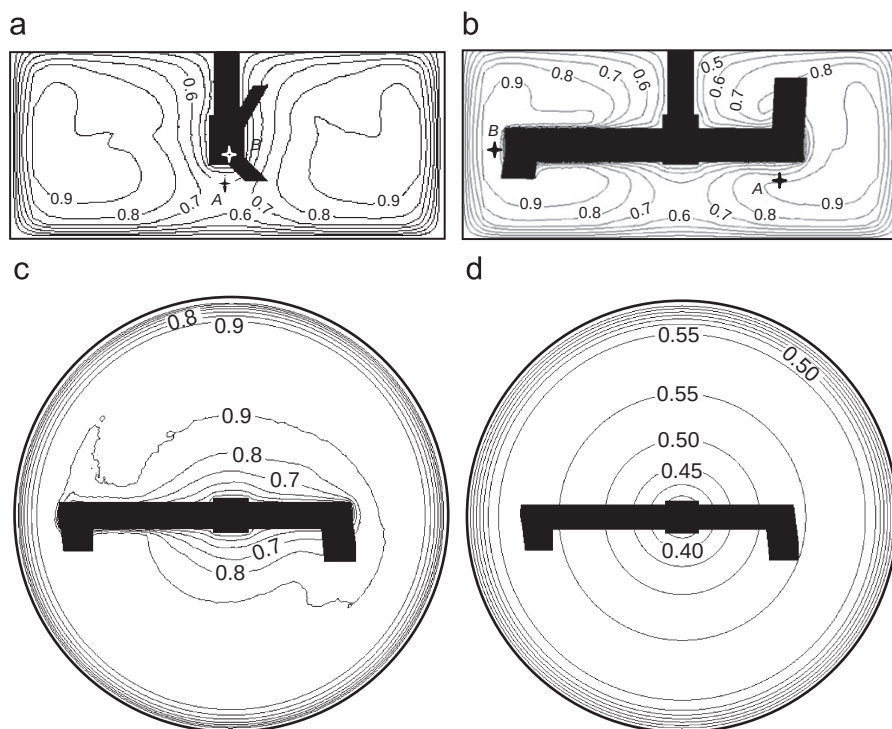


Fig. 8. Results obtained after coupling convergence, $t = 1000$ s. Dimensionless temperature distribution: (a, b) in vertical cross-sections; (c) in the horizontal plane through the impeller; and (d) at the free surface. $T_{\max} = 1620$ K; $N = 10$ rev min $^{-1}$; $P_{\text{tot}} = 60$ kW. The positions of monitoring points A and B (Fig. 6) are evident.

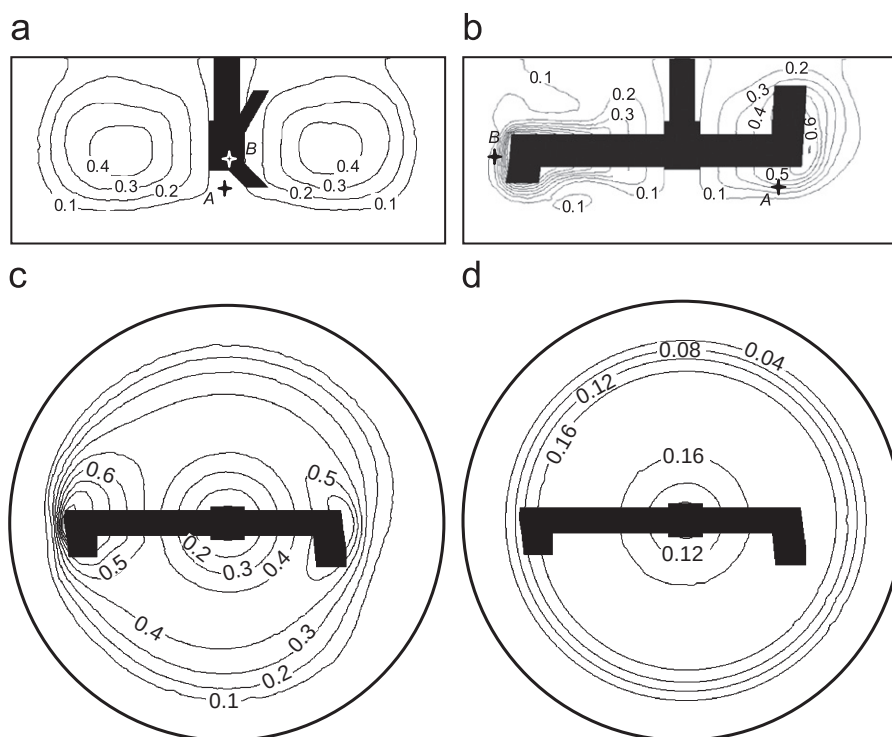


Fig. 9. Results obtained after coupling convergence, $t = 1000$ s. Flow pattern (a, b) in vertical cross-sections; (c) in the horizontal plane through the impeller; and (d) at the free surface. $V_{\max} = 0.22$ m s $^{-1}$; $N = 10$ rev min $^{-1}$; $P_{\text{tot}} = 60$ kW.

during experiments is well described by these values. The shape of the skull layer is presented in Fig. 7. The glass is assumed to be solidified when the viscosity is greater than 10^4 Pa s,

which corresponds to the viscosity at the flow point (Roscini, 1990). The thickness of the skull layer varies from a few millimeters to the centimeter scale. The solidified zone is thicker



Fig. 10. Dimensionless axisymmetric power source distribution obtained after coupling convergence, $t = 1000$ s. $Q_{th \max} = 6.64 \times 10^6 \text{ W m}^{-3}$; $N = 10 \text{ rev min}^{-1}$; $P_{\text{tot}} = 60 \text{ kW}$.

near the corner and the base because the effects of the stirrer are weaker in these zones.

Figs. 8 and 9 show temperature distributions and flow patterns in vertical cross-sections at the free surface and in the horizontal plane through the impeller. These results were obtained after coupling convergence after 1000 s, which corresponds to the thermal diffusion time within the glass bath. The maximum temperature is observed close to the crucible walls, where the power source distribution is highest. Fig. 10 shows the axisymmetric power source distribution obtained after convergence. The highest Joule power density is mainly located at the periphery of the melt. The differences between Joule power distributions at the start of the coupling (Fig. 5) and after convergence (Fig. 10) underline the necessity to perform coupled modeling. The highest Joule power density moves to the center of the bath during the coupling iterations and stretches along the crucible walls.

In addition, the calculated temperature distribution and flow pattern are quasi-axisymmetric within the glass bath. The fields are distorted only near the stirrer. This validates the possibility of performing 2D axisymmetric electromagnetic calculations.

4.2. Macroscopic study

In this section, mixing and heat transfer are considered.

4.2.1. Mixing efficiency

The power number N_p is commonly used as a macroscopic indicator to characterize stirrer efficiency. This number represents dimensionless power consumption

$$N_p = \frac{P}{\rho N^3 d^5}, \quad (13)$$

where N denotes the stirrer rotational speed (rev s^{-1}) and d is the impeller diameter. Power consumption P (W) may be calculated using two methods. The first integrates the viscous dissipation energy D_v into the whole bath volume (14). This calculation is representative of power consumption only under the hypothesis that the flow field reaches a steady state.

$$P = \int_V \mu D_v dV, \quad (14)$$

where

$$D_v = 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right] + \left[\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right]^2 + \left[\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right]^2 + \left[\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right]^2. \quad (15)$$

With the second method, power is calculated using the torque C applied to the impeller

$$P = 2\pi N C, \quad (16)$$

where

$$C = \left(\int_S (\vec{r} \times (\vec{\tau} \cdot \hat{n})) dS \right) \cdot \hat{a}. \quad (17)$$

Mixing in agitated vessels has been studied by several investigators. In all cases, the results for Newtonian fluids have been correlated using dimensionless equations of the general form

$$N_p = A Re^a Fr^b. \quad (18)$$

When deformation of the free surface is negligible, previous studies have demonstrated that N_p is almost constant in turbulent flows, whereas $N_p Re$ is constant in laminar flows (Nagata, 1975)

$$N_p = \frac{A}{Re}, \quad (19)$$

where

$$Re = \frac{\rho N d^2}{\mu}. \quad (20)$$

The constant A characterizes the system efficiency. If the mixing efficiency increases, the value of A will increase. Note that the value of A only depends on the stirred system, and remains constant regardless of the stirred fluid.

To obtain the value of A in the present study, numerical calculations were performed without considering the energy equation, with the fluid assumed to be isothermal. All the physical properties were assumed to be constant. Similar power consumption is obtained with both methods presented above. It is then possible to plot the classical mixing power curve (power number N_p vs. Reynolds number Re , Fig. 11). Very good agreement is observed between the computed and theoretical results, which confirms that the mixing regime is laminar. A value of 47 was obtained for the stirrer studied. Thus, the efficiency of this stirrer is quite low. In fact, its power consumption is equivalent to that of an impeller with three curved blades ($A=32$, Campolo et al., 2002), and is much lower than that of a high-efficiency impeller. For instance, the helical ribbon impeller has a value of 160 (Tanguy et al., 1997). It is easy to compute the value of A when the physical properties are constant. However, in the present study, the difficulty is in choosing a glass viscosity value to compute the value of the Reynolds number. Hence, the aim was to identify an equivalent viscosity so that $N_p Re$ may be constant. This method is adapted from the effective viscosity concept originally proposed by Metzner and Otto (1957) that is

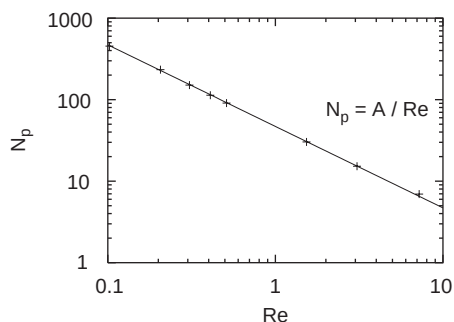


Fig. 11. Relation between power number N_p and Reynolds number Re for the stirrer studied with constant physical properties (isothermal conditions). $A = 47$.

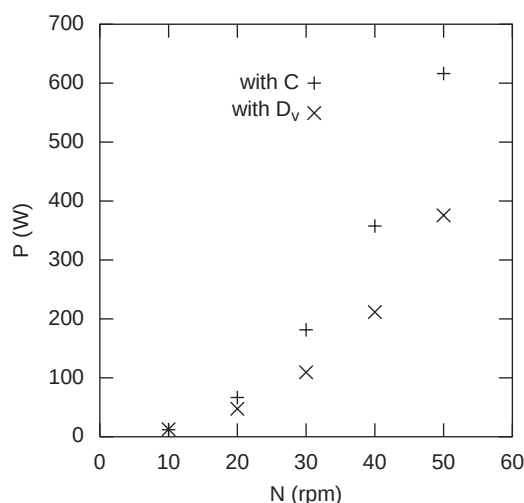


Fig. 12. Power consumption P calculated using two different methods vs. stirrer rotational speed N . $P_{\text{tot}} = 56 \text{ kW}$.

commonly used for non-Newtonian fluids. We performed calculations using the thermohydrodynamic and electromagnetic models presented above. Both methods for computation of P were tested. Fig. 12 compares the power consumption obtained with these two methods as a function of stirrer rotational speed. There is some discrepancy in the results that can be explained by the error in calculating the torque. In fact, the solidified glass layer implies that the normal component of the velocity gradient vector is equal to zero on the impeller walls. Thus, a method involving integration of the viscous dissipation energy was used in the present study. Knowing the value of the power consumption P allows us to calculate the value of the power number N_p . Then the equivalent Reynolds number Re_e can be computed from the mixing power curve $N_p(Re)$ (Fig. 11). We can use this law to study mixing within the glass bath, because this curve characterizes the stirred system. Finally, the equivalent viscosity μ_e is obtained from the equation

$$\mu_e = \frac{\rho N d^2}{Re_e}. \quad (21)$$

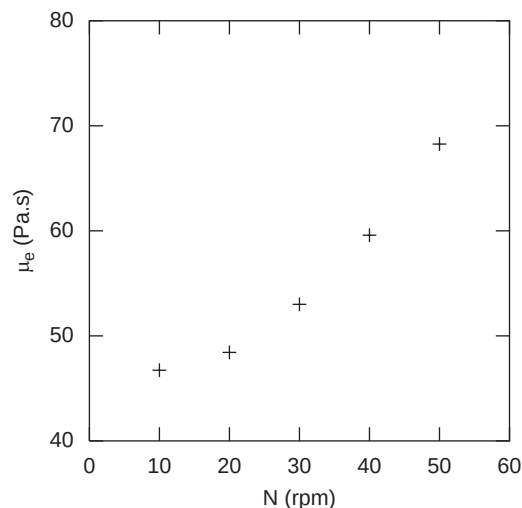


Fig. 13. Apparent viscosity μ_e vs. stirrer rotational speed N ; $P_{\text{tot}} = 56 \text{ kW}$.

This method is repeated for several angular velocity values. The equivalent viscosity is then plotted against the stirrer rotational speed (Fig. 13). It is evident that the equivalent viscosity increases more rapidly than the rotational speed. This can be explained by an increase in heat transfer on the cooled stirrer that leads to an increase in viscosity. Finally, according to the definition of the equivalent viscosity, the law $\mu_e(N)$ can be used to compute the Reynolds number so that $N_p Re$ is constant.

The method can be very useful in the optimization and operation of vitrification facilities. First, the method allows determination of the impeller power needed for a given angular velocity. In fact, the stirrer rotational speed leads to the equivalent viscosity and the power consumption can then be calculated from the mixing power curve. In addition, the torque on the stirrer leads to a value for the equivalent viscosity and then to an equivalent temperature using the thermal viscosity law (given in Fig. 2). This temperature is representative of the average temperature of the glass bath (Fig. 14). Consequently, the stirrer can be used to measure a representative temperature of the bath.

4.2.2. Heat transfer on the cooled stirrer walls

Heat transfer in agitated vessels has been studied less extensively than mixing over the past 50 years. Numerous heat transfer correlations have been proposed in the literature. The lack of a fundamental understanding of the flow fields in such vessels has forced an empirical approach, and generally the forced convective heat transfer characteristics have been expressed in the form

$$Nu = K Re^x Pr^y Vi^z, \quad (22)$$

where Pr is the Prandtl number and Vi is the viscosity correction term, defined as $Vi = \mu_{\text{wall}}/\mu_{\text{bulk}}$, where μ_{wall} denotes the viscosity close to the cooled (or heated) walls and μ_{bulk} is the viscosity of the bulk. The majority of investigations considered heat transfer from non-moving vessel walls. To date, few studies have identified correlations for heat transfer on a cooled

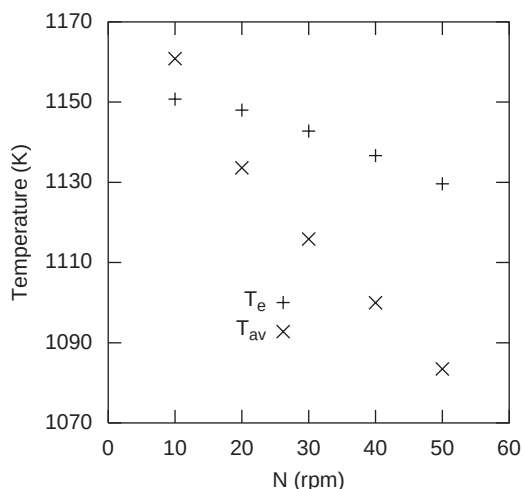


Fig. 14. Equivalent temperature T_e and average temperature of the glass bath T_{av} . $P_{tot} = 56$ kW.

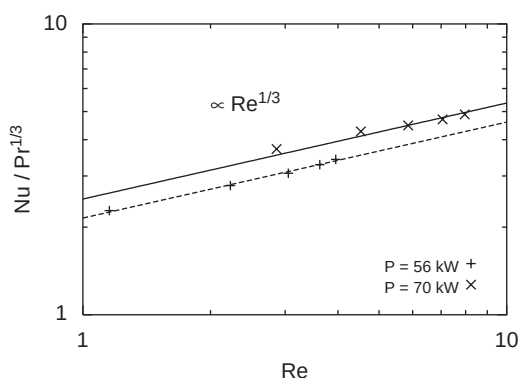


Fig. 15. Correlation between the Nusselt number and the Reynolds and Prandtl numbers for the total Joule power.

impeller. Nagata et al. (1972), Mitsuishi and Miyairi (1973) and Ishibashi et al. (1979) presented heat transfer correlations for helical-screw and helical-ribbon impellers. In all the cases (for moving or non-moving walls), for laminar regime researchers recommended a value of $1/3$ for the Reynolds number exponent x , $1/3$ for the Prandtl number exponent y and 0.14 for the viscosity ratio exponent z . The value of the constant K depends on the type of impeller.

In the present study, we obtained a value of $1/3$ for the Reynolds number and the Prandtl number exponents (Fig. 15). In addition, the viscosity correction term is no longer used in the heat transfer correlation when the Reynolds and Prandtl numbers are calculated using the equivalent viscosity μ_e . However, the value of K depends on the total Joule power P_{tot} injected within the glass bath. Finally, to obtain a unique constant for every total Joule power, we have to consider the ratio of heat transfer on the cooled stirrer to the total Joule power injected within the bath (Fig. 16). The value of this constant is $k = 0.005$ for this stirred system.

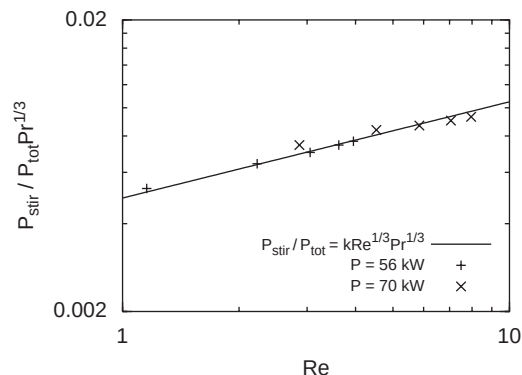


Fig. 16. Correlation between the heat transfer ratio for the cooled stirrer and the total Joule power injected within the bath. $k = 0.005$.

5. Conclusion

Electromagnetic, thermal and hydrodynamic phenomena have been modeled within a stirred molten-glass bath heated by direct induction. The study describes how coupling is achieved and what the simplifying assumptions are. This numerical modeling will provide valuable information for stirrer design and for the choice of location of measurement devices used to control the process. Secondly, a macroscopic study was carried out that will be very useful for characterizing flow fields and heat transfer within a stirred bath. Classical correlations commonly obtained for fluids with constant physical properties have been adapted for molten glass. This method will be very useful in the optimization and operation of vitrification facilities.

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